Response to Reviewers

Revision of: "Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis" (submitted to *Journal of Chemical Information and Modeling*)

Dear Editor,

We thank both reviewers for their careful, constructive reading of our manuscript. Reviewer 1 raised two major concerns and Reviewer 2 raised nine major points and seven minor points. We have revised the manuscript (V8) to address every point. Below we respond point by point; locations refer to the revised main text (M) or Supporting Information (SI).

Reviewer 1

R1.1 — Uneven structural evidence quality across the four targets

Reviewer: "The four targets have vastly different levels of structural validation. These differences would influence the 'target breadth' result."

Response: We agree, and we now make the gradient of structural confidence explicit rather than implicit. The Introduction states that the panel spans "a gradient of structural confidence — PfDHFR carries a co-crystallized inhibitor, PfCRT is represented by a proxy cavity, and PfClpP and PfATP4 by mechanism-motivated regions without co-crystallized small-molecule ligands — so target breadth is read as a pattern of target-specific evidence, not as four equivalent demonstrations of biological engagement" (M, Section 1). The Methods (Section 2.3) describe each anchor's evidence class, and Section S9 tabulates the four evidence classes (catalytic-site ligand, Y01 cavity proxy, catalytic triad, P-type ATPase machinery). Accordingly, we report all four-target results target by target and never compute a cross-target composite, so the heterogeneity affects interpretation (breadth as a hypothesis) rather than a pooled number.

R1.2 — Validation strategy: feasibility and retrospective validation

Reviewer: "The proposed validation strategy lists five categories of experiments but provides no preliminary data, no estimated feasibility, and no prioritization beyond naming three candidates. At minimum a retrospective validation against known antimalarial scaffolds would strengthen the workflow's credibility."

Response: We strengthened the strategy in three ways.

1. **Retrospective validation.** We docked five approved antimalarials (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) against the PfDHFR wild-type/mutant panel (N51I,

C59R, S108N, I164L; wild-type receptor stripped of the co-crystallised MTX to avoid retained-ligand bias, consistent with the mutant preparation) and the corrected PfCRT panel (LYS-76 WT, K76T, K76A) under the declared protocols (40 runs, exhaustiveness 32), and computed per-drug RRS profiles to test whether the framework recovers clinically documented resistance signatures. The result is honest-negative and is reported as such (Table S7; Section 4.5): pyrimethamine RRS is ≈100% across all four PfDHFR mutants (N51I 100.0, C59R 100.0, S108N 99.7, I164L 100.0), chloroquine RRS is 99.6% (K76T) and 98.8% (K76A) on PfCRT—both within the pipeline-null spread of the corrected channel (99.4–100.5%)—and artemisinin/cipargamin show PfDHFR RRS ≈113–118% while chloroquine/lumefantrine are PfDHFR non-binders (consistent with their non-antifolate mechanisms). The framework therefore does not recover the clinical pyrimethamine/chloroquine resistance signatures under the corrected protocols, which further bounds the interpretation of RRS as a resistance predictor rather than a biological measurement.

2. **Prioritization.** Section 4.5 now gives an explicit, ordered five-step experimental sequence for the five top-priority candidates (PP-15, PP-05, PP-06, PP-11, PP-13): IC_{50} against wild-type parasites and recombinant proteins → isogenic mutant-line assays (the exact N51I/C59R/S108N/I164L and K76T/K76A panels used here) → orthogonal binding (SPR/ITC) → MD refinement → ADMET/synthetic-accessibility assessment.

3. **Feasibility framing.** We now state explicitly that the IC_{50} and mutant-panel assays are the standard first steps for computational hit triage and require only the five candidates plus two isogenic panels, and that all claims remain computational hypotheses until those measurements exist.

Reviewer 2

We thank Reviewer 2 for the encouraging framing and for the specific, actionable list. Each point is addressed below.

R2.1 — Unreported N_fav–RRS correlation; power limitation; "demonstrate"

Reviewer: Spearman $\rho(N_{\text{fav}}, \text{RRS}) = +0.515$, $p = 0.034$ should be reported; the PNS–RRS $p = 0.020$ "trend" and the "demonstrate" language are in tension with §2.5; with $n = 17$ the power to detect $\rho = 0.5$ at $\alpha = 0.017$ is only 0.37.

Response: The N_fav–RRS correlation is now reported in Section 3.6 and Section S8 (Table Table S9), and was additionally assessed by a 100,000-permutation test. Because the corrected PfCRT re-dock (R2.3) changed the PfCRT column and the favorability definition (R2.2), the final numbers on the corrected matrix are $\rho = +0.714$ ($p = 0.0013$; permutation $p = 0.0019$), significant at the Bonferroni threshold $\alpha = 0.017$. The Discussion now reads the cross-metric findings with the power limitation stated explicitly: at $n = 17$, a two-sided Spearman test at $\alpha = 0.05$ reaches 80% power only at $|\rho| \geq 0.62$, and at the Bonferroni-corrected $\alpha = 0.017$ the power to detect a correlation of $\rho = 0.5$ is only ≈ 0.37 (the reviewer's estimate), so only moderate-magnitude associations are detectable at this sample size (Section 4.1; Section S8). The same caveat qualifies the PNS–RRS and RRS–wild-type-score associations, which are also reported on the corrected matrix (PNS–RRS $\rho = -0.714$, $p = 0.0013$; RRS–mean $|S_{WT}|$ $\rho = +0.691$, $p = 0.0021$; both significant at $\alpha = 0.017$). As an additional robustness check, the companion P2 layer reports that the PNS–RRS association survives and strengthens after conditioning on molecular weight and Murcko-scaffold prevalence (partial $\rho = -0.6154$ versus raw $\rho = -0.2098$, $n = 12$ complete-two-target set), consistent with the negative PNS–RRS direction reported here and indicating that the association is not a size or scaffold artifact.

R2.2 — The -6.0 kcal/mol threshold performs the cross-target comparison

Reviewer: A single cutoff passes 7/17 (PfDHFR) but 12/17 (PfCRT/PfATP4); favourability should be defined within each target (rank/percentile) and Table 2 recomputed.

Response: We adopted the reviewer's within-target definition (option b). Favorability is now defined per target as a Vina estimate at or below that target's cohort median (PfDHFR -5.92, PfCRT -8.06, PfClpP -6.02, PfATP4 -6.38 kcal mol^{-1} on the corrected matrix; Table S8), and Table 2 and the N_fav–RRS statistics were recomputed accordingly (Section 2.5, Section 3.5; Section S8). Under the per-target-median definition, $\rho(N_{\text{fav}}, \text{RRS}_{\text{mean}}) = +0.714$ ($p = 0.0013$; permutation $p = 0.0019$, 100,000 draws), significant at $\alpha = 0.017$. For transparency, the retired single-cutoff variant gave $\rho = +0.515$ ($p = 0.034$); the within-target definition is the one reported because it avoids a fixed cross-target cutoff, as the reviewer requested.

R2.3 — PfCRT 6UKJ isoform and grid box (critical)

Reviewer: 6UKJ is the 7G8 isoform; residue 76 is threonine (K76T); the 28 Å box excludes residue 76; the PfCRT panel is the sole resilience source for PP-02/PP-06/PP-11/PP-13 and contributes to PP-01/PP-15; anchor to the central negatively charged cavity and construct a 3D7-like wild-type reference.

Response: We agree this was the most serious flaw. We thank the reviewer for the precise geometric analysis. Three corrections are in place:

1. **3D7-like wild-type receptor.** The deposited 6UKJ carries threonine at residue 76; we restored lysine to obtain a 3D7-like wild-type receptor (LYS-76) and use it as the PfCRT wild-type reference (Section 2.3; Section S12.1).
2. **Grid anchored to the drug-interaction cavity.** The corrected grid is centered at $(152.99, 151.042, 159.379)$ with 25 Å dimensions, the V2-corrected box used in our resistance-resilience analysis. P2Rank pocket prediction independently validates the box: the top-ranked pocket (score 162.7, probability 0.999) lies 5.7 Å from the grid center (Section S12.1), i.e., the box covers the central cavity, not a peripheral region.
3. **Re-dock completed.** All 17 candidates were re-docked against the corrected LYS-76 receptor (and, for the R2.5 control, the K76T, K76A, and pipeline-null receptors) on the V2 grid (exhaustiveness 32): 68 runs, all finite. Table S1 (PfCRT column), Table 1 (PfCRT RRS values with the corrected wild-type denominator), and Table 2 (N_fav per-target- median assignments) were regenerated from these runs. The corrected wild-type scores are substantially more favorable (mean -8.42 vs -6.19 kcal mol^{-1}), and the corrected PfCRT RRS values all fall between 99.1 and 100.6%. Class counts changed from 6/5/5/1 to 7/4/6/0 (A*/B/C/D), and the corrected RRS range is 73.2–115.6%.

The corrected-receptor protocol is the one described in the manuscript, and all result tables were regenerated from the corrected runs (scripts `p1_r23_pfcrt_redock.py` and `p1_r23_rrs_recompute.py` in the repository).

R2.4 — DEKOIS worse-than-random; retained MTX; Hany et al.

Reviewer: EF@1% = EF@5% = 0.00 suggests retained MTX blocking the antifolate pocket; re-run on a ligand-stripped receptor; Hany et al. (Drug Des. Devel. Ther. 2025, 10.2147/DDDT.S537065) applied DEKOIS 2.0 to WT and quadruple-mutant PfDHFR with Vina and rescoring.

Response: We report EF@1% and EF@5% (both 0.00) alongside the ROC-AUC in Table S11 and discuss the retained-MTX hypothesis explicitly: the MTX redocking failure (RMSD $\approx$ 30 Å) and the PfDHFR grid position indicate that the null enrichment partly reflects grid positioning rather than scoring-function failure alone (Section 2.3.1; Section S12.2). We now cite and discuss Hany et al. as the closest prior work: they applied the same DEKOIS 2.0 protocol to wild-type and quadruple-mutant PfDHFR with AutoDock Vina and showed that machine-learning rescoring (CNN-Score, RF-Score-VS) moved the benchmark from worse-than-random to better-than-random, corroborating our conclusion that enrichment depends on consensus or rescoring rather than raw Vina ranking (Section S12.2).

We performed the requested re-run as a controlled two-arm experiment on the identical DEKOIS panel (the same 40 actives and 1200 property-matched decoys, same grid): (i) the deposited 7F3Y receptor with MTX retained, which reproduces the original baseline under our pipeline (ROC-AUC 0.502 [95% CI: 0.423–0.586], EF@1% 0.00, EF@5% 0.50), and (ii) an MTX-stripped receptor rebuilt in the same coordinate frame (ROC-AUC 0.563 [95% CI: 0.477–0.646], EF@1% 0.00, EF@5% 1.00), both at exhaustiveness 32 (the revised-protocol standard, identical in both arms so the comparison is internally controlled). The result is honest-negative for the blockade hypothesis: the confidence intervals overlap and EF@1% remains zero in both arms, so removing MTX yields only a marginal, non-significant improvement (ΔAUC = +0.060) and does not restore enrichment. We therefore report that the near-chance DEKOIS result is attributable to the single-method Vina scoring function and grid positioning (near the NADPH cofactor site) rather than to steric blockade by the retained cofactor ligand, and the manuscript now states this explicitly with the two-arm data added to Table S11 and Section S12.2 (Section 2.3.1; Section 4.5).

R2.5 — Batch effect: WT and mutants prepared by different routes

Reviewer: Provide the required control: WT processed through the mutant pipeline, use the resulting distribution as a null, place class boundaries outside it; explain the six RRS values above 100%.

Response: We now explain the values above 100% in Section 3.4: they arise when the mutant docking estimate is more favorable (more negative) than the wild-type estimate on the same receptor; they are scoring-function ratios and do not imply biological gain-of-function. The requested null control was run: the corrected LYS-76 wild-type receptor was passed through the identical mutation-preparation pipeline without applying any mutation (pipeline-null), and all 17 candidates were docked against this null receptor and against the K76T and K76A mutants on the same V2 grid (Section S12.7). The null RRS distribution (null ratios vs the corrected wild-type denominator) spans 99.4–100.5% (mean 100.0%); the K76T ratios span 99.1–100.6% and K76A 99.4–100.4%. The maximum deviation of any mutant ratio from its null ratio is 1.2 percentage points (K76T) and 0.5 (K76A), both far below the 10-point class boundary spacing. The 80%/70% class boundaries therefore fall inside the null spread on the corrected PfCRT channel, and we do not assert PfCRT-based class distinctions beyond the null spread: PfCRT RRS values near 100% are reported as neutral-within-null. The receptor- preparation contribution to the old PfCRT signal is thus quantified as the entire difference: the old spread (70.4–110.2% across K76T/K76A) reflected the raw 6UKJ (7G8, K76T) receptor used as "wild type" and a grid that excluded the drug-interaction cavity.

The neutral-within-null reading is corroborated by two independent companion analyses in the P2 layer. (i) An external docking replication (39 ligands × 8 PfDHFR/PfCRT states, 312/312 finite records, prepared entirely independently of the present panel) gives a bootstrap RRS distribution of 100.45 [99.59, 101.32], so the corrected PfCRT channel's null spread (99.4–100.5%) falls inside the pipeline noise of a fully separate preparation. (ii) In the companion P2 layer, RRS class assignments on its canonical panel are robust to score perturbation: under bounded ±1 kcal/mol perturbations of individual docking scores (272 candidate–score records), the class was unchanged in 212/272 (77.9%), and leave-one-mutant-out and wild-type-threshold (4 vs 5 kcal/mol) sensitivity runs leave its reported class distributions stable. Separately, an MD-based MM-GBSA $\Delta\Delta G$ analysis in the companion layer found 0 of 12 PfDHFR/PfCRT mutants significantly weaker and 1 significantly tighter at 95% CI (max $|\Delta\Delta G|$ = 5.37 kcal/mol), corroborating that mutant-to-wild-type ratios above 100% reflect retention rather than gain.

R2.6 — Repository returns 404; data unavailable

Reviewer: The repository URL returns 404; the wild-type denominators do not appear elsewhere; provide a maintained repository with license, release tag, and archival DOI.

Response: The repository is now publicly accessible at `https://github.com/NanaEngo/Malaria_codesV2` under the MIT license (`LICENSE` at the repository root), with a versioned archival release `v1.0.0` (`https://github.com/NanaEngo/Malaria_codesV2/releases/tag/v1.0.0`) containing the complete revision codebase, the canonical result directories (including the two-arm DEKOIS re-run of R2.4), and a checksummed staging package for archival (`zenodo_package_P1/`, sha256-verified manifest). A permanent Zenodo DOI is being minted for this release and will be inserted into the Data Availability statement (main) and the SI reproducibility section (Section S11) before resubmission. In parallel, the WT/mutant score table required to recompute Table 1 is now included directly in the SI (Table S6: the PfDHFR wild-type/mutant panel and the corrected PfCRT LYS-76/K76T/K76A scores underlying every RRS denominator), so the RRS denominators no longer reside only in the repository.

R2.7 — PNS and ACSI never defined

Response: Section 2.5 now gives the formal definitions. ACSI $= 0.40 \times D_{\text{DrugBank}} + 0.25 \times D_{\text{ANPDB}} + 0.20 \times f_{sp^3} + 0.15 \times \text{NPL}$ with min–max normalization over the 17-candidate cohort (D = minimum Tanimoto distance to approved-drug / African-NP reference sets, f_{sp^3} = fraction of sp^3 carbons, NPL = natural-product-likeness). PNS is computed from a STRING protein–protein interaction network at confidence threshold 700, with PfCRT centrality imputed as the network mean (0.151); Table S10 reports the imputation sensitivity (Spearman $\rho = 0.963$–1.000 across zero to double the imputed value). No PNS or ACSI value is interpreted causally.

R2.8 — Table S12 vs Section S12.3 contradiction

Reviewer: §12.2 says no pose-reproduction rate is claimed for PfClpP, yet Table S8 lists "1 success" for all four targets; 9N10 has no small-molecule ligand; the score-stratified MMV rows are circular.

Response: The validation tables were rebuilt to remove the contradiction. Tables S12 and S14 now separate (i) full-ligand redocking (PfDHFR P218, RMSD 1.42 Å; PfCRT Y01 proxy, 1.78 Å) from (ii) fragment pose reproduction (PfClpP peptide, 1.65 Å) and cofactor pose reproduction (PfATP4 ADP, 1.23 Å), and state explicitly that (ii) is not equivalent to full-ligand redocking and that 9N10 contains no co-crystallized small-molecule ligand. The MMV rows for PfCRT/PfATP4 are explicitly labelled score-stratified retrodictive consistency checks, not external discrimination. The text in Section S12.3 was revised to match.

R2.9 — Single seed; PP-15 marginally above threshold

Reviewer: PP-15 exceeds the cutoff by 0.045 (PfDHFR) and 0.084 (PfCRT); run at least five seeds and report dispersion.

Response: Five-seed sensitivity runs (seeds 101–505) are reported in Table S13 for PP-15 and PP-01 on PfDHFR and PfCRT; mode-1 affinities are stable to seed choice (maximum spread 0.054 kcal/mol). The table transparently states that those runs used the P2 canonical-grid protocol and therefore report seed dispersion for that protocol. After the R2.3 re-dock on the corrected receptor and declared grids, the per-target-median favorability assignments (R2.2) place 48 of 68 candidate–target pairs at least 0.3 kcal mol^{-1} from the relevant target median; the remaining 20 pairs lie within 0.3 kcal mol^{-1}, including four that sit exactly at the median (PP-05 PfATP4, PP-10 PfClpP, PP-12 PfCRT, PP-13 PfDHFR), so the within-target favorability counts should be read with this boundary sensitivity in mind. The corrected-protocol single-run scores used for those assignments are the same values reported in the corrected Table S1, and the re-dock used a single seed with exhaustiveness 32 as in the original protocol. The seed dispersion shown in Table S13 (< 0.06 kcal mol^{-1}) was measured under the P2 canonical-grid protocol and therefore provides a scale reference for seed sensitivity rather than a direct bound on the corrected single-seed V2-grid runs; the boundary analysis above (20 pairs within 0.3 kcal mol^{-1} of a target median, four exactly at the median) identifies which N_fav assignments are most sensitive to single-seed variation.

R2m — Minor points

1. **Duplicated paragraph (pp. 17–18):** the verbatim duplicate is removed; the top-candidate summary now appears once in Section 3.5 and is not repeated verbatim in the Discussion.

2. **VAE citation:** we now cite Gómez-Bombarelli et al. (ACS Cent. Sci. 2018) for the SMILES VAE (Section 2.1).

3. **"99.3% reduction":** reframed as an absolute reduction (263,424 → 1,936 calculations) with the explicit caveat that this describes computational cost, not active-compound retention, and that a retention experiment using known actives would be needed (Section 4.4).

4. **RRS eligibility notation:** all instances now use $|S_{\mathrm{Vina},i,\mathrm{WT},t}|$ consistently (Section 2.4; Section S7); no ΔG notation remains.

5. **Section S12 numbered twice; decoy counts:** SI now contains a single S12 block (S12.1–S12.7); the DEKOIS decoy count is consistently 1,200.

6. **Figure 2 annotations:** the figure was regenerated with ρ, p, and n annotated in each panel from the corrected canonical data (PNS–RRS $\rho = -0.714$, $p = 0.0013$; ACSI–RRS $\rho = -0.190$, $p = 0.465$; $n = 17$ in both panels).

7. **MPO sensitivity deferred:** the essential results are now summarized in Section S10.2 (25 weight perturbations; Spearman $\rho = 0.792$ for the top-1000 ranking), so the manuscript stands alone while the full methodology remains in the companion chemical-space manuscript.

Status of items formerly marked [TO COMPLETE]

Completed for this revision: (i) corrected-receptor PfCRT re-dock of the 17-candidate panel and regeneration of Table S1/Table 1/Table 2 (R2.3/R2.9); (ii) the batch-effect null control and class-boundary placement (R2.5; Section S12.7); (iii) Figure 2 statistical annotations (R2m.6); (iv) per-target-median favorability and recomputed Table 2/N_fav–RRS (R2.2); (v) retrospective validation against five approved antimalarials (R1.2; 40 runs, honest-negative result reported in Table S7 and Section 4.5); (vi) the WT/mutant score table in the SI (R2.6; Table S6); and (vii) the controlled two-arm MTX-retained versus MTX-stripped DEKOIS re-run (R2.4; honest-negative result, two rows added to Table S11 and discussed in Section S12.2). Additionally executed for R2.6: the repository is now public (MIT license, release tag `v1.0.0`, GitHub release), with the staged Zenodo package ready; only the Zenodo DOI minting (author account action) remains before resubmission.

Sincerely,
Myke Vital Sao Temgoua, Jean-Pierre Tchapet Njafa, Penabei Samafou,
Wilfred Fon Mbacham, Serge Guy Nana Engo